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Contributions of Muscle Forces to Stability and Mobility in Radiography-driven Models of Total Shoulder Arthroplasty

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ABSTRACT

Several complications are associated with total shoulder arthroplasty (TSA), including glenoid loosening and joint instability. These complications are linked to imbalances in glenohumeral muscle forces, which cause eccentric loading that occurs on the outer edge of the glenoid component, rather than in the center of the glenoid, and can be damaging to the implant and destabilizing to the joint. The purpose of this study was to develop musculoskeletal models of TSA patients using their skeletal geometry acquired from CT and their kinematics acquired with high-speed stereo radiography (HSSR), to predict glenohumeral joint loading and muscle forces for abduction and flexion, and describe how these loads differ from their native contralateral side. CT data were collected for both shoulders of 6 TSA patients (3 anatomic, 3 reverse). Models were developed using CT reconstructions of each patient's scapula, humerus, and placed implant geometry with CT derived alignment. Mesh morphing was used to create personalized muscle attachment sites on the scapula. Simulation of abduction and flexion to quantify muscle and joint forces was driven with limb-specific HSSR kinematics. At 90° abduction, reverse TSA increased superior shear force by 11.2% bodyweight (BW) and decreased medial compressive force by 27.5 %BW compared to each patient's contralateral shoulder. Lateral deltoid elevation moment arms and stability ratios increased 9.61 mm and 0.58, respectively. At 90° abduction, anatomic TSA patients increased humeral internal rotation 52.7° and decreased rotator cuff force by 9.4 %BW. Patient-specific musculoskeletal models quantified differences in glenohumeral joint mechanics. This study revealed muscle and kinematic mechanisms

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that could lead to edge loading and glenoid loosening in TSA and can inform personalized surgical planning for implant alignment with the goal of improving patient outcomes.

INTRODUCTION

The number of patients receiving total shoulder arthroplasty (TSA) has risen in the past decade and was predicted to increase 235% in annual volume from 2017 to 2025 (350,558 cases/year) [1]. The two most common forms of TSA are anatomic TSA (aTSA), which mimics the natural anatomy of the glenohumeral joint (GHJ), and reverse TSA (rTSA), which reverses the ball-and-socket relationship of the native GHJ. While TSA frequently relieves pain and restores mobility, several complications persist, including loosening of the glenoid component and joint instability [2], [3], [4], [5]. Glenoid component loosening is defined as implant migration, and is often followed by glenohumeral instability, which is associated with increased pain, decreased function, and eventual revision surgery [5], [6]. One hypothesized cause of glenoid component loosening is decreased joint compression and increased shear loading from the muscles (Yanagawa 2008, Fig. 1 [7]). This can lead to eccentric loading, which occurs when the humeral head migrates such that the face of the glenoid component experiences force toward its edge, rather than in the center, and can be damaging to the implant and destabilizing to the joint [5], [6], [8]. Researchers have hypothesized that TSA alters muscle moment arms and lines of action, which influences muscle force production and resulting joint loading [9], [10], [11], [12], [13]. These prior simulations of shoulder mechanics, however, used generic models that don’t account for individualized muscle

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lines of action or joint center locations based on patient bone and implant geometry, which may influence the magnitude and direction of simulated muscle forces. Moreover, prior studies used generic kinematics, which do not account for the specific axial orientation and plane of elevation of each patient's humerus, which also may influence the magnitude and direction of muscle forces and resulting joint forces.

The lateral deltoid is the prime mover during abduction and flexion tasks in the native shoulder, and its force vector during a motion dictates much of the GHJ reaction load vector [7], [9], [10], [14]. As such, altering its relation to the GHJ center due to TSA is likely to influence the amount of shear and compression loading at the joint. TSA has been shown to significantly alter the joint center of rotation, which changes the mobilizing and stabilizing potential of the surrounding musculature [9], [15]. A more medial and inferior positioning of the joint center location in rTSA relative to the native joint center may allow the lateral deltoid to more easily elevate the humerus via an increased moment arm during abduction and flexion [9], [10], [11], [12], [13], [15], [16], [17]; however, it may also align the muscle more superiorly, increasing the potential to impart shear force on the glenoid and decreasing the ability to stabilize the joint [18]. Ackland et al [9] predicted that average GHJ force was slightly increased after aTSA (2.6 %BW increase) during abduction. They also predicted significantly decreased average GHJ force after rTSA (32.3 %BW decrease), along with decreased average compressive joint loading (45.4 %BW decrease) and slightly increased average superior shear force (4.7 %BW increase).

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While there are no current methods to noninvasively measure muscle forces and joint reaction loads *in vivo*, computational modeling has emerged as the standard method to predict muscle forces and joint loading. Still, the majority of computational modeling studies use scaled versions of generic musculoskeletal models [8], [9], [14], [15] which may fail to capture how individual variability in bone morphology and muscle attachment site locations influence muscle and joint reaction forces. Additionally, many studies utilize generic kinematic profiles [9] or kinematics captured from skin-based optical motion capture [14], which is known to poorly measure motion of the scapula [19]. Moreover, shoulder kinematics are highly individualized and change with age [20], [21], [22], and therefore models using accurate, patient-specific *in vivo* kinematics are preferred. To our knowledge, there are no studies that have combined 1) stereo radiography measurements of patient-specific shoulder movements, 2) patient-specific glenohumeral alignment from CT imaging, and 3) patient-specific scapular morphology and muscle attachment sites in a musculoskeletal model to predict muscle forces and resulting joint reaction loads at the glenohumeral joint for TSA populations.

The primary objective of this study was to develop patient-specific musculoskeletal models accounting for individualized patient shoulder morphology and motion for both the implanted and contralateral shoulders for six different TSA patients (three aTSA, three rTSA) to evaluate differences in GHJ loading and muscle forces between each patient's implanted and native shoulders at 90° abduction and flexion. Further, we aimed to quantify muscle moment arms, stability ratios, and 3D kinematics

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to explore how changes to joint geometry impacted the mobilizing and stabilizing potential of the musculature.

2. METHODS

2.1 *In vivo* Data Collection

Upper limb experiments were performed on six patients (five male, one female), mean age: 76±5.4 years (range: 71-84 years), mean mass: 100±15.3 (range: 74.8-117.4 kg), mean height: 1.71±0.88 m (1.58-1.79m) who had previously undergone TSA (three aTSA and three rTSA). Bilateral-upper extremity CT scans (512 x 512 x 0.625 mm resolution) were taken of each participant's implanted and native contralateral limbs. Patients performed two shoulder tasks while being recorded by high-speed stereo radiography (HSSR) and marker-based optical motion capture as part of a larger study [23], [24], [25]. Each patient performed the two functional tasks with their right and left upper limb: coronal plane abduction and sagittal plane flexion (Fig. 1). Patients began with their arm at their side, elevated their arm to their maximum range of motion, and were instructed to keep their trunk steady and their thumb pointing up and palm facing forward/to the side during the motions. All patients provided informed consent to a protocol approved by the University of Denver IRB 1066258.

The HSSR imaging system [26] was used to quantify shoulder kinematics using system settings described by Hamilton et al [24] with the exception that sampling rates were collected at 25 Hz. Glenohumeral kinematics derived from HSSR provided 3D relative bone motion that avoids the errors associated with surface marker motion

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capture [28] and reliably measures kinematics to <1° of rotation [26], [29]. Simultaneous marker-based motion capture (Vicon, Centennial, CO) recorded the position of reflective markers placed on the thorax, which was not captured by HSSR. Marker data was used to develop a thorax coordinate system, which allowed for calculation of transformation matrices describing scapulothoracic kinematics. Patient-specific bone geometries were segmented from the CT scan consisting of the proximal right and left humerus and the right and left clavicle and scapula (Simpleware ScanIP, Synopsys, Sunnyvale, USA). Bony landmarks were manually identified for each bone and assigned a local coordinate system based on ISB recommendations [30]. A fixed thorax position was defined by constructing a coordinate system using optical motion capture marker data of each patient in their neutral pose at the beginning of each activity. Scapular and humeral bone geometries were tracked in two imaging planes of the HSSR to accurately determine patient-specific joint movement in 3D space (Autoscoper, Brown University). Tracking involved manipulating the virtual position of the three-dimensional (3D) bone and implant models in space until the two-dimensional projections of the bones onto each camera view created the resulting captured image [31], [32], [33]. Transformation matrices describing the position and orientation of the thorax, scapula, and humerus body segments were extracted, and then kinematics of the bones were calculated for all six degrees-of-freedom (DOF) at the glenohumeral and scapulothoracic joint with a third-order one-dimensional median filter (*medfilt1(x)*, MATLAB r2022a, MathWorks, Natick, MA) applied to all trajectories. Euler angle decompositions for the scapulothoracic (YXZ) and glenohumeral (YXY) joints were performed in accordance with

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ISB recommendations [30] to match the rotational sequence of the shoulder model developed in OpenSim 4.1 (see below).

2.2 Musculoskeletal Model of the Shoulder

Shoulder musculoskeletal models (Fig. 2) of the six patients were created in OpenSim [34] by adapting a model created by Wu et al [14]. Patient-specific bone geometry of the scapula, clavicle, and proximal humerus replaced the bone geometry in the original model. The glenohumeral joint representation from Wu et al. was retained and consisted of a 3DOF joint located at the geometric center of the humeral head. A 6DOF scapulothoracic joint was implemented to allow for direct prescription of the position and orientation of the scapula in relation to the thorax. Additionally, the sternoclavicular joint was replaced with a point constraint relating the proximal end of the clavicle to the superolateral portion of the sternum. Thorax body segments were scaled to the patients based on superior-inferior measurements of the sternum and medial-lateral measurements of the ribcage from scout CT images. Segmentations of the humerus from CT did not include the distal third of the humerus. Thus, scaling factors were calculated using the acromion and lateral epicondyle optical markers to determine the length of each patient's humerus and define the position of the elbow joint center.

All 26 (13 glenohumeral and 13 scapulothoracic) muscle-tendon units from the model created by Wu et al. [14] were included in the model of each shoulder. In the model created by Wu et al. [14], the locations of muscle origins on the scapula were based on a prior upper extremity model of the Visible Human Male [35]. For each

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patient-specific model, a non-linear morphing algorithm, based on a generalized regression neural network (GRNN) developed in our prior work [36], was used to morph the attachment sites from the Visible Human Male scapula to that of the patients to accurately produce individualized muscle attachment sites that maintained anatomic relevance [37] (Fig. 2). For each patient, humeral insertion sites for glenohumeral muscles were determined by locating anatomical landmarks on the humerus model of each patient [38]. Wrapping objects created by Wu et al. (2016) were updated for each patient. All floating via points were removed from the original model and replaced with cylindrical wrapping objects for the deltoid and rotator cuff muscles to create more realistic muscle paths [39]. Updated muscle-tendon unit (MTU) lengths were verified to ensure that adding wrapping objects did not significantly alter each muscle's ability to generate force when compared to MTU lengths using via points (See Supplemental Material, Supp. Fig. 1). Thorax wrapping objects were scaled based on scout CT images of each patient. Small manual adjustments to final wrapping object locations and sizes were made to ensure the muscle paths wrapped over the bones, reducing muscle-bone intersections during patient-specific movements (Fig. 2).

Consistent with the shoulder model of Wu et al. [14], a Hill-type model of muscle force was used with patient-specific parameter adjustments. Maximum isometric force was scaled based on the whole-body mass of each patient in relation to the whole-body mass of the subject in the original model. Tendon slack length and optimal fiber length were also scaled for each patient model based on the length of the patient humerus segment in relation to that of the original humerus segment to preserve their ratio as

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defined in the original model. For each patient model, muscle properties were the same in their implanted and contralateral sides. Full thickness supraspinatus tears were simulated in all shoulders with rTSA by deactivating the muscle unit in all static optimization simulations, as described by Ackland et al. [9]. Patient-specific models used in our analysis are available at https://simtk.org/projects/ps_tsa_modeling.

2.3 Simulation of Abduction and Flexion

All patient simulations were performed in OpenSim using the MATLAB API 4.1. Six DOF scapulothoracic and 3 DOF glenohumeral kinematics from HSSR measurements were used as input to static optimization simulations. The muscle physiology tag within the static optimization tool was used to account for muscle force-length-velocity relations. The static optimization minimized the sum of squares of muscle activations [7], [34], [40]. No reserve actuation was necessary for convergence at the glenohumeral joint.

The MATLAB OpenSim API was used to extract glenohumeral joint reaction force, muscle forces, muscle moment arms, and muscle stability ratios over the full movement cycle for each patient and trial. Results for each patient were then evaluated at a discrete pose (90° thoracohumeral elevation) during abduction and flexion. Patient joint reaction loads were validated against experimentally measured loading data from OrthoLoad for both abduction and flexion [41] (Fig. 3). Muscle stability ratios quantify a muscle's potential to dislocate the shoulder joint along the degrees of freedom specified in the calculation. Superior stability ratios greater than one indicate that the muscle will exert more superior, destabilizing force than medial, compressive force. Individual

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muscle superior stability ratios were calculated as described by Yanagawa et al [7] by dividing the superior shear component of the muscle line of action by the medial, compressive component for each frame during abduction and flexion. Additionally, average stability ratios were quantified for a full movement cycle, as described by Ackland et al [42]. An OpenSim muscle force direction plugin [43] was used to calculate all glenohumeral effective muscle lines of action in the scapula coordinate system. All aTSA and rTSA group comparisons were made by quantifying the difference between each patient's implanted shoulder and their own contralateral shoulder, then calculating the average difference among aTSA and rTSA patients, effectively treating the contralateral limb as each patient's control. After analyses, glenohumeral axial orientation for all trials was quantified using a XZY decomposition sequence and validated against calculated true axial rotation values [44]. At the final frame of abduction, the change in axial rotation values from each patient's starting pose were comparable between the true and YZX calculation methods (true: 30.9±24.1°, YZX: 30.4±29.1°) (See Supplemental Material, Supp. Fig. 2). All data processing was performed in MATLAB.

3. RESULTS

3.1 Glenohumeral Joint Reaction Force Validation

The glenohumeral joint reaction forces (GHJRF) of all patients' implanted shoulders were similar in magnitude and trend during abduction and flexion to joint forces reported for instrumented patients who had received aTSA, with joint forces increasing with increasing elevation (Fig. 3) [41]. At 90° of elevation, instrumented

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patient joint forces were 665.4±197.1N (abduction) and 553.5±180.3N (flexion). At the same pose, simulated aTSA joint forces were 641.5±127.2N (abduction) and 551.8±174.4N (flexion), and simulated rTSA joint forces were 504.4±121.1N (abduction) and 426±81.6N (flexion). All simulated mean forces at 90° elevation for abduction and flexion fall within the 5-95 confidence interval of instrumented forces (abduction: 503.1, 827.9N, flexion: 356.6, 750.6N).

3.2 Patient Kinematics

Patient kinematics were similar between the implanted and contralateral sides during abduction and flexion except for notable differences in range of motion and humeral internal rotation (See Supplemental Material, Supp. Fig. 3).

rTSA:

During abduction and flexion, all rTSA patients achieved less maximum humeral elevation in their implanted limb than in their contralateral native limb (abduction: -22.7±27.4 ° flexion: -17.1±5.4°). Implanted shoulders exhibited less internal rotation compared to their contralateral shoulder, although there was considerable variation across all patients at 90° of arm elevation (-37.4±51.6 degrees and -13.9±27.8 degrees during abduction and flexion, respectively).

aTSA:

During abduction, all aTSA patients exhibited maximum humeral elevation within 10 degrees of their contralateral limb (mean difference: 1.04 ±9.2°) but achieved on average 14.5 ±15.6° less humeral elevation in flexion. Implanted shoulders exhibited

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more internal rotation (or decreased external rotation) compared to their contralateral shoulders at 90° of arm elevation (52.7±23.3° and 44.5±37.0° during abduction and flexion, respectively).

3.3 Muscle Forces

Muscle activation profiles during abduction compared favorably in magnitude and trend with previously published electromyography data [7]. Specifically, activation of the anterior and middle deltoids increased with increasing elevation, the posterior deltoid was relatively dormant, and the infraspinatus and supraspinatus muscles were consistently active during the motion (See Supplemental Material, Supp. Fig. 4).

During abduction and flexion, the lateral deltoid, anterior deltoid, and infraspinatus muscles produced the most force, although variability existed between patients. (Fig. 5). Specifically, at 90° abduction and flexion, the roles of the infraspinatus, subscapularis, and clavicular pectoralis major muscles were variable, as described below and in the Supplemental Material (See Supplemental Material, Supp. Fig. 5).

rTSA:

At 90° abduction, on average, lateral deltoid force increased 13.0±6.4 %BW, anterior deltoid force decreased 10.3±7.6 %BW, and subscapularis force decreased 12.6±2.9 %BW compared to their contralateral shoulders (Table 1). The combined rotator cuff muscle group forces decreased 24.5±13.3 %BW (Table 1). At 90 degrees of flexion, on average, infraspinatus force decreased 6.8±1.4 %BW, and combined rotator cuff muscle group forces decreased 11.2±3.6 %BW compared to contralateral (Table 2).

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Infraspinatus muscle forces were greater than 5 %BW in two of three rTSA shoulders at 90° abduction and one of three rTSA shoulders at 90° flexion. Clavicular pectoralis major forces were greater than 5 %BW in two of three rTSA shoulders at 90° abduction and one of three rTSA shoulders at 90° flexion.

aTSA:

At 90° of abduction, lateral deltoid force increased 7.03±6.4 %BW, anterior deltoid force decreased 7.30±5.8 %BW, clavicular pectoralis major force increased 7.05±5.79 %BW, and infraspinatus force decreased 8.60±3.6 %BW compared to their contralateral shoulders (Fig. 5). The combined rotator cuff muscle group forces decreased 9.4±3.3 %BW in (Table 1) compared to contralateral. At 90 degrees of flexion, on average, infraspinatus force decreased 6.75±6.1 %BW compared to contralateral (Table 2).

Infraspinatus muscle forces were greater than 5 %BW in two of three aTSA shoulders at 90° abduction. Clavicular pectoralis major forces were greater than 5 %BW in two of three aTSA shoulders at 90° abduction. Subscapularis forces were greater than 5 %BW in two of three aTSA shoulders at 90° abduction one of three at 90° flexion.

3.4 Deltoid Muscle Moment Arms and Stability Ratios

rTSA

At 90° of abduction, the lateral deltoid elevation moment arms increased 9.61 mm, and the anterior deltoid moment arms increased 11.10 mm (Fig. 6). Axial rotation

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moment arms about the long axis of the humerus, which describe the ability of the muscles to generate internal/external rotation, were comparable to their intact side. Average superior stability ratios increased in the anterior deltoid (0.27 increase) and lateral deltoid (0.58 increase) during abduction when compared to their contralateral shoulder (Fig. 7), indicating potential to generate greater shear force relative to compression.

aTSA

Deltoid muscle elevation moment arms were comparable to those on the contralateral side (Fig. 6). However, at 90° of abduction, axial rotation moment arms for the anterior deltoid increased 6.71 mm and lateral deltoid increased 5.80 mm, meaning these muscles had more ability to generate internal rotation torque. Superior stability ratios were similar between implanted and contralateral shoulders (Fig. 7).

3.5 Joint Reaction Forces

Notable trends existed in the comparison of three-dimensional joint forces between the implanted and contralateral limbs within the rTSA cohort during abduction and flexion, as well as substantial variability in the comparison of these three-dimensional joint forces (See Supplemental Material, Supp. Table 1), specifically in the anterior direction, within both the aTSA and rTSA cohorts, which will be further explained below.

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rTSA:

At 90° of abduction, on average, superior shear forces increased 11.2±2.2 %BW, medial force decreased 27.5±7.2 %BW, and joint force magnitude decreased 21.0±8.3 %BW. (Fig. 4). At 90° of flexion, superior shear forces increased 9.5±8.7 %BW, medial forces decreased 20.0±17.2 %BW, and joint force magnitude decreased 12.2±11.8 %BW (Fig. 4).

Changes in anterior loading between implanted and contralateral limbs in individual patients were variable. For example, at 90° abduction, anterior force was 21.8 %BW greater in patient TSA03, while anterior force was lesser in patients TSA04 and TSA06 (11.3 and 5.1 %BW, respectively) when compared to their own contralateral native limbs. At 90° flexion, anterior force was greater in patients TSA03 and TSA04(5.0 and 13.6 %BW, respectively), while anterior force was lesser (5.8 %BW) in patient TSA06 when compared to their own contralateral limb.

aTSA:

No trends in differences in joint reaction forces between the implanted and contralateral limbs across aTSA patients were detected, although there were notable differences between individual patient's implanted and contralateral limbs during abduction and flexion. Specifically, at 90° abduction, anterior and medial compressive force was lesser in patient TSA01 (-14.9, -13.7 %BW respectively), and superior force was greater (15.2 %BW) when compared to their own contralateral limb. Conversely, anterior and medial force was greater in patient TSA05 (8.2, 7.6 %BW respectively), and along with superior force was lesser (-9.4 %BW) when compared to their own

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contralateral shoulder. At 90° flexion, anterior and medial compressive force was again lesser in patient TSA01 (-1.4, -21.1 %BW respectively), while anterior, superior, and medial compressive forces were lesser in patient TSA05 (-12.5, -19.1, -11.7 %BW, respectively), and all three force components were greater in TSA02 (2.2, 7.7, 10.1 %BW respectively) when compared to their own contralateral limb (Supp. Fig. 6).

4. DISCUSSION

Patient-specific and limb-specific musculoskeletal models of the implanted and contralateral native shoulders for a cohort of six TSA patients were developed in OpenSim. Glenohumeral joint force and muscle forces were calculated using static optimization with prescribed patient-specific scapulothoracic and glenohumeral kinematics for abduction and flexion acquired with stereo radiography. Patient-specific geometry and kinematics produced unique muscle and joint forces, supporting the benefits of analyzing TSA mechanics with a patient-specific approach.

Maximum joint reaction forces during abduction in the simulated contralateral native, aTSA, and rTSA shoulders were comparable to those measured with instrumetned implants (OrthoLoad database, Fig. 3). Notably, rTSA joint forces were, on average, smaller than simulated aTSA and experimental aTSA (OrthoLoad) joint forces. This trend of decreased rTSA joint forces when compared to aTSA joint forces aligns with results from Ackland et al [9] using generic kinematic profiles. However, there are several important differences between the native, aTSA, and rTSA joint force predictions in the study by Ackland et al [9] and the present study. Anterior-posterior joint forces in

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the present study are smaller (less posterior) in native, aTSA, and rTSA shoulder types (-8.2±22.7, -4.9±17.1, -7.5±25.3 %BW, respectively) than in the prior study (-30.4±0.1, -24.7±4.0, -13.6±4.7 %BW, respectively). Another notable difference is that standard deviations of maximum anterior joint force in all native, aTSA, and rTSA models (22.7, 17.1, 25.3 %BW, respectively) were much larger in the present study than those from Ackland et al [9] (0.1, 4.0, 4.7 %BW, respectively). The smaller standard deviation values in anterior joint forces in Ackland et al. [9] may result from the use of a generic kinematic profile, which may cause muscle force predictions to follow a similar activation pattern and result in similar joint reaction loads. In the present study, the differences in kinematics between an individual patient's implanted and contralateral shoulder, specifically in terms of axial orientation, may contribute to a wider range of muscle activation strategies to achieve these patient-specific motions (Fig. 8), which ultimately influence three-dimensional joint reaction force predictions. Increased internal rotation of the humerus in implanted shoulders relative to each patient's own contralateral shoulder was accompanied by decreased deltoid internal rotation torque and decreased rotator cuff external rotation torque relative to their native side (Fig. 8).

Comparisons of joint reaction forces between implanted and contralateral shoulders for each aTSA and rTSA patient were made at 90° abduction and flexion because joint forces were consistently larger at this elevation than at lower elevations (Fig. 4), which may pose a greater risk of off-center, damaging glenoid loading and joint instability. At these potentially high-risk poses, variability existed in the differences in anterior loading at 90° elevation between implanted and contralateral shoulders in both

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aTSA (abduction mean difference: 2.2 ±11.7 %BW, flexion mean difference: 3.7±7.7 %BW) and rTSA (abduction mean difference: 1.8±17.6 %BW, flexion mean difference: 4.2±9.7 %BW) patients. One aTSA patient and two rTSA patients demonstrated more posteriorly directed (decreased anterior) force at 90° abduction, while the others demonstrated more anteriorly directed (increased anterior) forces when compared to their contralateral native shoulders. At 90° flexion, two aTSA patients and one rTSA patient resulted in greater posteriorly directed force when compared to their contralateral shoulders, while the others experienced greater anteriorly directed force. This concept that patients with the same implant type (aTSA or rTSA) experience different changes in anterior joint forces when compared to their own contralateral limbs suggests that accurate personalized kinematics, in concert with limb-specific model development, may be important considerations for accurately predicting how the anterior loading component will differ for a specific aTSA or rTSA patient from their healthy shoulder during elevation tasks. Given that the rocking horse phenomenon is driven by off-center loading [5], [6], quantifying anterior-posterior loading on a patient-specific basis, particularly at higher angles of elevation where joint forces are larger, may be useful for predicting the risk of glenoid loosening for a given surgical alignment.

One of the guiding principles of rTSA is that medializing the joint center will lead to larger deltoid moment arms, making it easier for patients to elevate their arms in the presence of rotator cuff deficiency [10]. Indeed, all rTSA patients had increased anterior and lateral deltoid elevation moment arms in this study (Fig. 7). However, while anterior deltoid force decreased, lateral deltoid force increased in all rTSA shoulders in

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comparison to native shoulders at 90° abduction and flexion (Table 1). Simulated supraspinatus tears may contribute to the increased lateral deltoid force since the supraspinatus normally contributes to elevation of the humerus [8], [45]. Antagonist muscle force from the subscapularis muscle decreased on average as well, which may contribute to increased shear forces. Indeed, at 90° humeral elevation, when compared to each patient's own contralateral shoulder, rTSA shoulders on average experienced greater superior shear force (abduction mean increase: 11.2 ±2.2 %BW, flexion mean increase: 9.5±8.7 %BW) and less medial, compressive force (mean abduction decrease: 27.5±7.2 %BW, mean flexion decrease: 20.0±17.2 %BW), indicating a higher risk for large glenoid base-plate stresses and potential for joint subluxation [9].

Results supported that aTSA more closely reproduces the native mechanics of the shoulder than rTSA, particularly when the rotator cuff is intact (Fig. 4b) [46]. While muscle force differences between the native and implanted shoulders were fewer and lesser in magnitude in aTSA compared to the rTSA cohort during both abduction and flexion (Table 1), some notable trends emerged. When compared to each patient's own contralateral limb, aTSA shoulders produced increased lateral deltoid force (7.03±6.39 %BW) and decreased anterior deltoid and infraspinatus force (7.30±5.84 %BW), 8.60±3.61 %BW, respectively) (Table 1). The infraspinatus is well positioned to create joint compression [7], thus a potential consequence of the decrease in force produced by this muscle in aTSA shoulders, in conjunction with greater lateral deltoid force, may increase the likelihood of off-center loading of the glenoid, which can contribute to glenoid loosening [5].

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Muscle force and GHJRF changes within the rTSA cohort were the result of altered muscle lines of action in relation to the joint center due to TSA position and patient-specific kinematics. In all patients, the lateral deltoid had greater average stability ratios than the anterior deltoid (Fig 7), meaning it was more aligned to develop superior shear force at the glenoid. However, some patients chose arm motion strategies that favored lateral deltoid elevation torque, potentially due to its larger maximum isometric force and elevation moment arm (Fig 7). This caused increased shear on the glenoid and a decrease in compression, particularly in rTSA shoulders, where the lateral deltoid was more superiorly aligned than in the contralateral shoulder (Fig 7). Ensuring that the placement of an arthroplasty will allow a patient to utilize the anterior, lateral, and posterior heads of the deltoid muscle may be an important consideration for increasing joint stability and longevity of the implant, particularly in rTSA patients.

Given that changes in joint center were minimal in the aTSA cohort, differences in muscle force predictions may be more strongly influenced by arm orientation during each movement. Interestingly, each aTSA patient exhibited a more internally-rotated (or less externally-rotated) humerus during abduction and flexion (Fig. 8). The orientation of the humerus reduced total deltoid internal rotation moment arms (Fig. 7), and consequently, deltoid internal rotation torque (Fig. 8), which may explain why less infraspinatus force was necessary to maintain dynamic equilibrium in these shoulders (Table 1). While it was not explicitly analyzed in this study, force-couple relationships between deltoid and rotator cuff muscle moment arms are likely to also exist in the

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transverse plane. Since rotator cuff activation is crucial for glenohumeral stability and balanced glenoid loading [42], it is important to understand how it is influenced by patient-specific arm orientation during voluntary movement. Future studies should explicitly address the impact of glenohumeral plane and axial orientation on muscle-driven joint loading through sensitivity analyses.

There were limitations to the analyses. First, since the functional status of each rotator cuff muscle in each shoulder of the patients was unknown, only full thickness supraspinatus tears were simulated in rTSA patients [9]. Including patient-specific measures of infraspinatus and subscapularis function would further personalize GHJRF predictions [8]. Second, muscle insertions and origins were identified through morphing, scaling, and bony landmarks. Identifying precise deltoid attachment sites via MR imaging might improve the patient-specificity of these models. Third, maximum isometric force of the muscles was scaled to body weight. Performing measurement of maximum isometric efforts for the patients might have produced better estimations of patient-specific muscle parameters in the models. Lastly, models built for contralateral shoulders were assumed to be healthy, which may not be the case in older populations where shoulder pathologies and rotator cuff tears often persist on both sides [47].

The musculoskeletal models developed in this study discount glenohumeral translations by assuming a ball-and-socket GHJ, which may influence muscle and joint reaction force predictions [39]. However, humeral head translations for abduction and flexion have been reported to be small (1.4, 2.4 mm, respectively) [29], [39], [48], [49], suggesting that translations would have little influence the muscle moment arm and

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stability ratio calculations. Additionally, these models provide predictions of muscle forces that are unique to each patient given their joint geometry and motion patterns, but they do not predict contact mechanics between the arthroplasty articular surfaces or fixation surfaces. Nevertheless, the patient-specific muscle-driven joint reaction forces calculated in this study can be applied to patient-specific deformable finite element (FE) models to predict pressure contours on articulating and fixation surfaces of the implant [8]. This framework would allow for analysis of how the forces exerted by a patient's muscles interact with their specific implant, alignment, and fixation, which may be used to inform how surgical decisions may impact implant stresses on a patient-specific basis.

In conclusion, we found unique joint loading and muscle forces in all shoulders, with substantial increases in superior shear (abduction: 11.2±2.2 %BW, flexion: 9.5±8.7 %BW) and decreases in compressive loading (abduction: -27.5±7.2 %BW, flexion: -20.0±17.2 %BW) after rTSA. Crucially, compared to the contralateral shoulders, rTSA increased deltoid elevation moment arms and superior stability ratios; however, individual variation in kinematics and joint geometry created variation in internal rotation moment arms, leading to differences in anterior-posterior joint loading. By utilizing patient-specific musculoskeletal models combined with patient-specific kinematics measured by HSSR, this work enabled analyses that included the influence of individuals' arm orientation on muscle-driven joint loading, providing more specificity to explanations of muscle force contributions to glenoid loading and the mechanisms leading to glenoid loosening. These types of models can help clinicians make informed

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decisions about optimal implant placement for patients, potentially improving TSA outcomes.

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DISCLOSURES

The authors have no conflicts of interest related to this work.

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NOMENCLATURE

aTSA	Anatomic total shoulder arthroplasty
rTSA	Reverse total shoulder arthroplasty
%BW	Percentage of bodyweight
AP	Anteroposterior
SI	Superoinferior
ML	Mediolateral

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Figure Captions List

Fig. 1 The two dynamic activities of the study: coronal plane abduction with palm facing anteriorly (top, left) and sagittal plane flexion with palm facing medially (top, right) B) HSSR imaging of a patient performing abduction (bottom, left) and flexion (bottom, right)

Fig. 2 Figure 2: OpenSim musculoskeletal models showing scapula muscle attachment sites, marked by red X's (top), visualizations of full models with muscle sets (middle), and visualizations of full models with GH joint centers (bottom). (A) The baseline model developed by Wu et al. (2016) with original muscle attachment sites and wrapping objects, leading to instances of muscle-bone intersections (B) One of the twelve limb-specific models from this study via morphing, scaling, and manual attachment site adjustment to reflect patient-specific bone morphology and fewer muscle-bone intersections.

Fig. 3 Measured glenohumeral joint force magnitudes from all available subjects in the Orthoload database who completed abduction and flexion to 90° humeral elevation compared to all simulated aTSA and rTSA joint reaction force magnitudes from this study. A) Abduction (top) and flexion (bottom) joint reaction forces plotted against thoracohumeral elevation. B) Mean

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and standard deviation joint reaction forces at 90° abduction (top) and flexion (bottom).

Fig. 4 (A) Joint reaction force curves for the implanted and contralateral shoulders of all patients plotted against thoracohumeral elevation during abduction and flexion (positive values represent anterior, superior, and medial force, respectively) (B) Box plots of differences in joint reaction force components between each patient's implanted and contralateral limb for each implant type at 90 degrees of thoracohumeral elevation (positive values indicate increased anterior, superior, and medial force, respectively). Forces are expressed as acting on the glenoid in the scapular coordinate system.

Fig. 5 Muscle force curves for the implanted and contralateral shoulders for all patients calculated from static optimization for the main contributing glenohumeral muscles plotted against thoracohumeral elevation during (A) abduction and (B) flexion.

Fig. 6 (A) Glenohumeral elevation and axial rotation moment arms of the deltoid muscles in all implanted and contralateral shoulders plotted against thoracohumeral elevation during abduction (positive values represent elevation moment arms and internal rotation moment arms) (B) Box plots displaying differences in glenohumeral elevation and axial rotation moment arms for the deltoid muscles between each patient's implanted

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Fig. 7 (A) Average stability ratios of the deltoid muscles for the implanted and contralateral shoulders of all patients plotted against thoracohumeral elevation during abduction (positive values indicate superior stability ratios; negative values indicate inferior stability ratios) (B) Box plots displaying differences in average stability ratios of the deltoid muscles between each patient's implanted and contralateral shoulder for each implant type (positive values indicate increased superior stability ratios, or more superior inclination). Average stability ratios were calculated as described by Ackland and Pandy [42].

Fig. 8 Glenohumeral orientation of the implanted and contralateral limbs of the three aTSA patients in the study plotted against the percentage of the complete abduction motion (left); sum of the axial rotation torque produced by the anterior, lateral, and posterior deltoid muscles plotted against humeral elevation (middle); sum of the axial rotation torque produced by the infraspinatus, subscapularis, supraspinatus, and teres minor muscles plotted against humeral elevation (right) during (A) abduction and (B) flexion. Positive values indicate internal axial

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Table Caption List

Table 1 Differences in muscle group, as well as individual muscle forces between the implanted and contralateral limbs for each group (aTSA, top; rTSA, bottom) at 90° of abduction. Positive values indicate increased force in the implanted limb compared to the contralateral limb.

Table 2 Differences in muscle groups and muscle forces between the native and implanted limbs for each group (aTSA, top; rTSA, bottom) at 90° of flexion. Positive values indicate increased force in the implanted limb compared to the contralateral limb.

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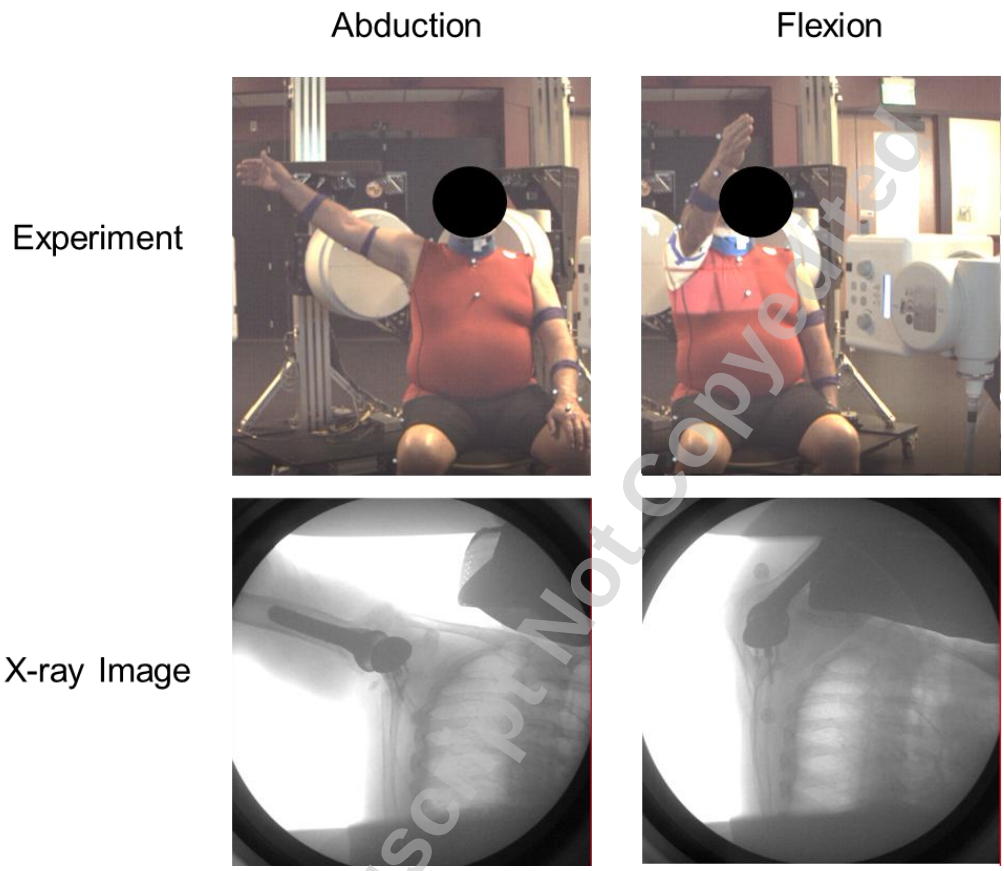
Figure 1:



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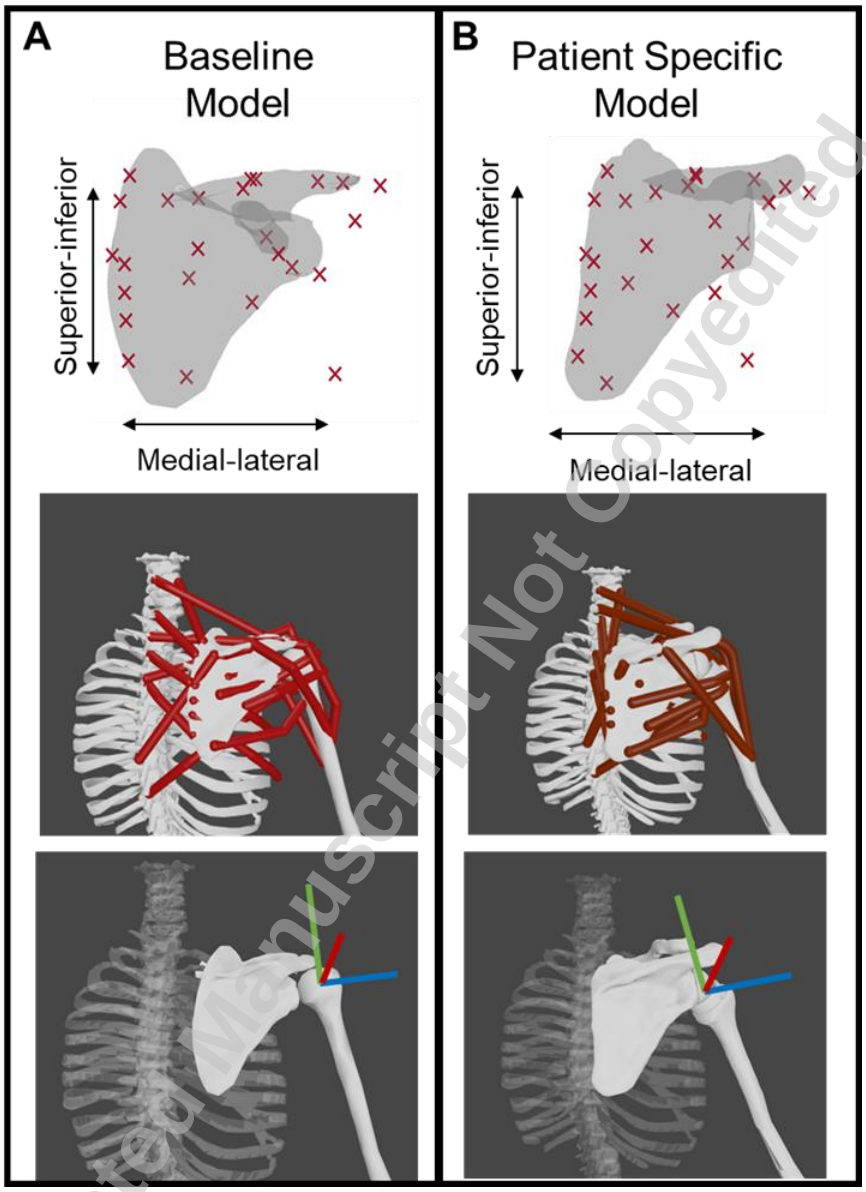

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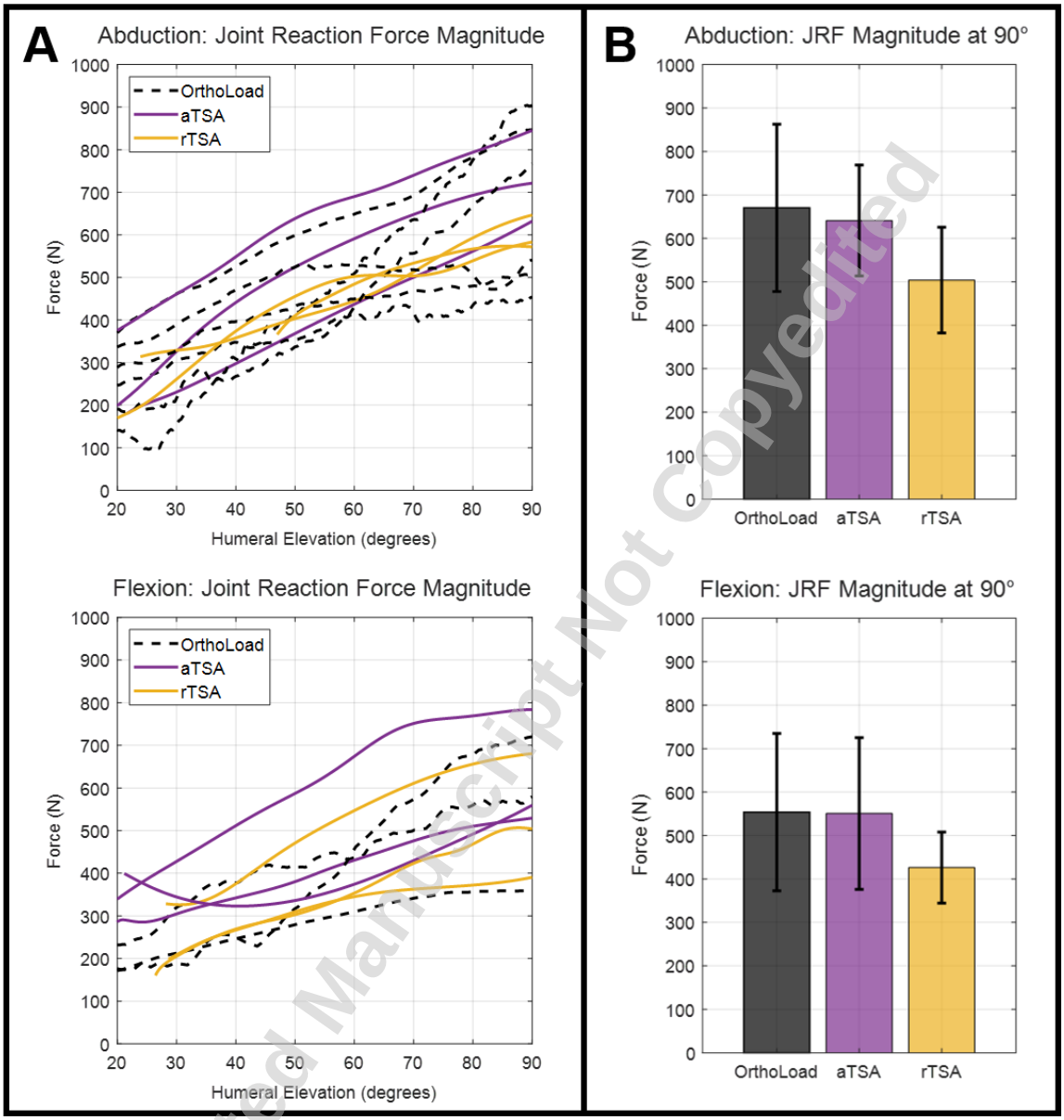
Figure 3



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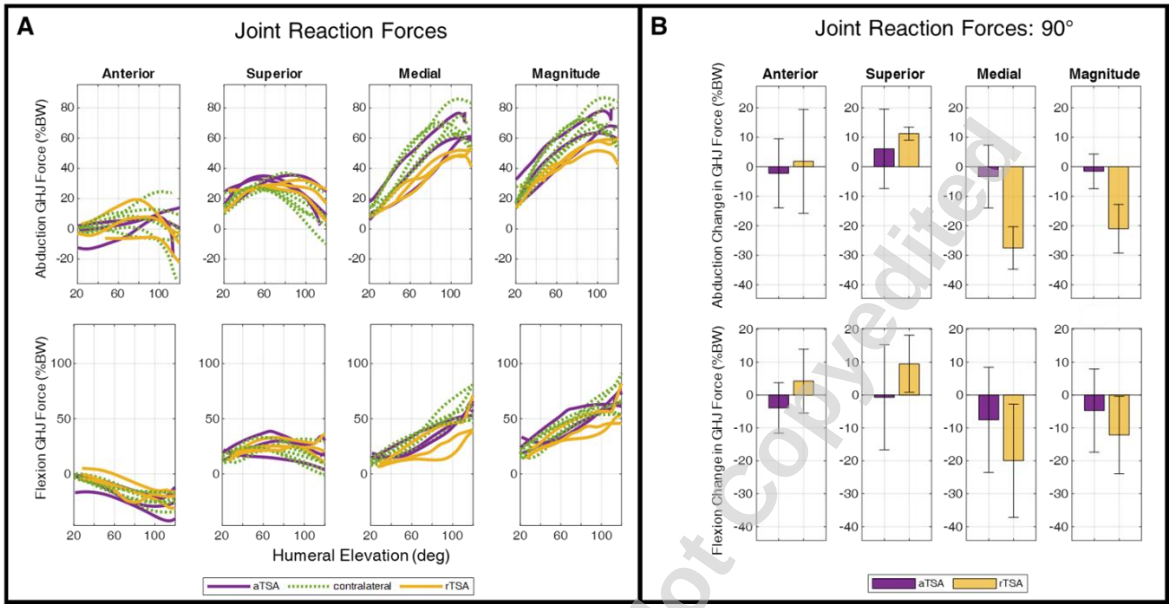

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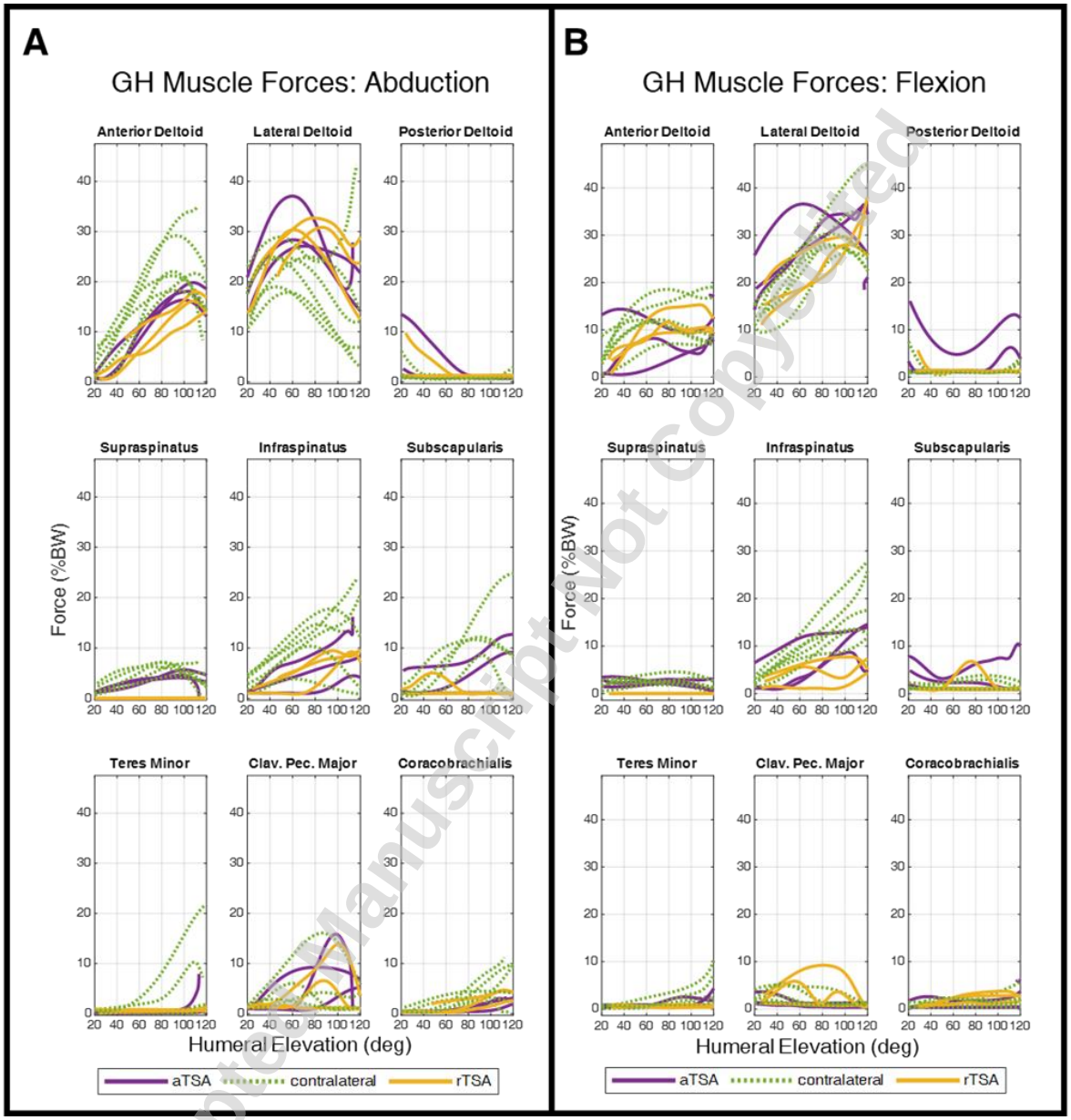

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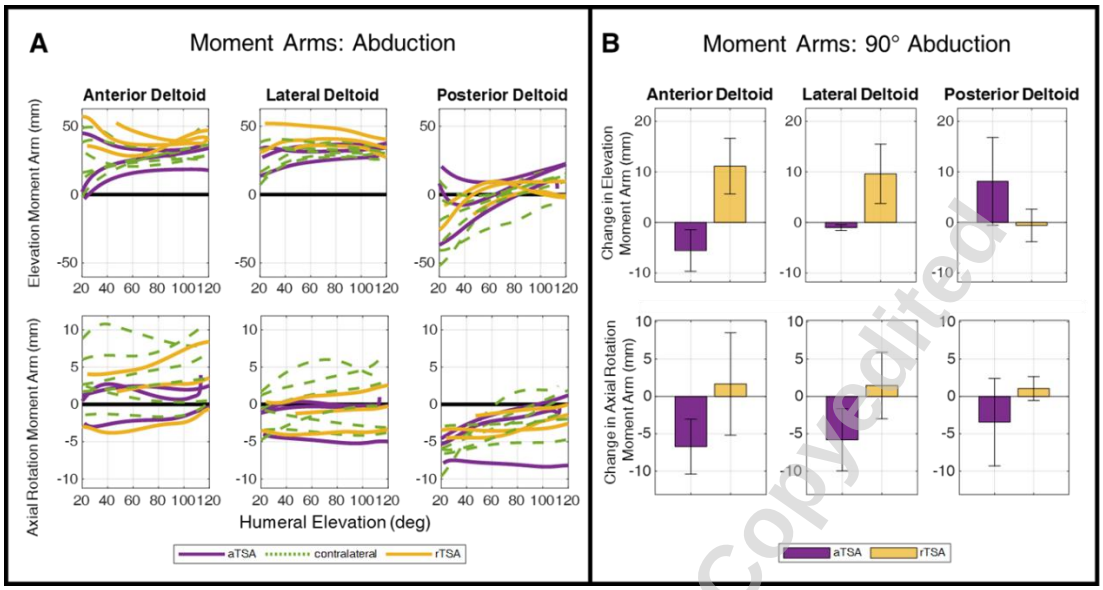

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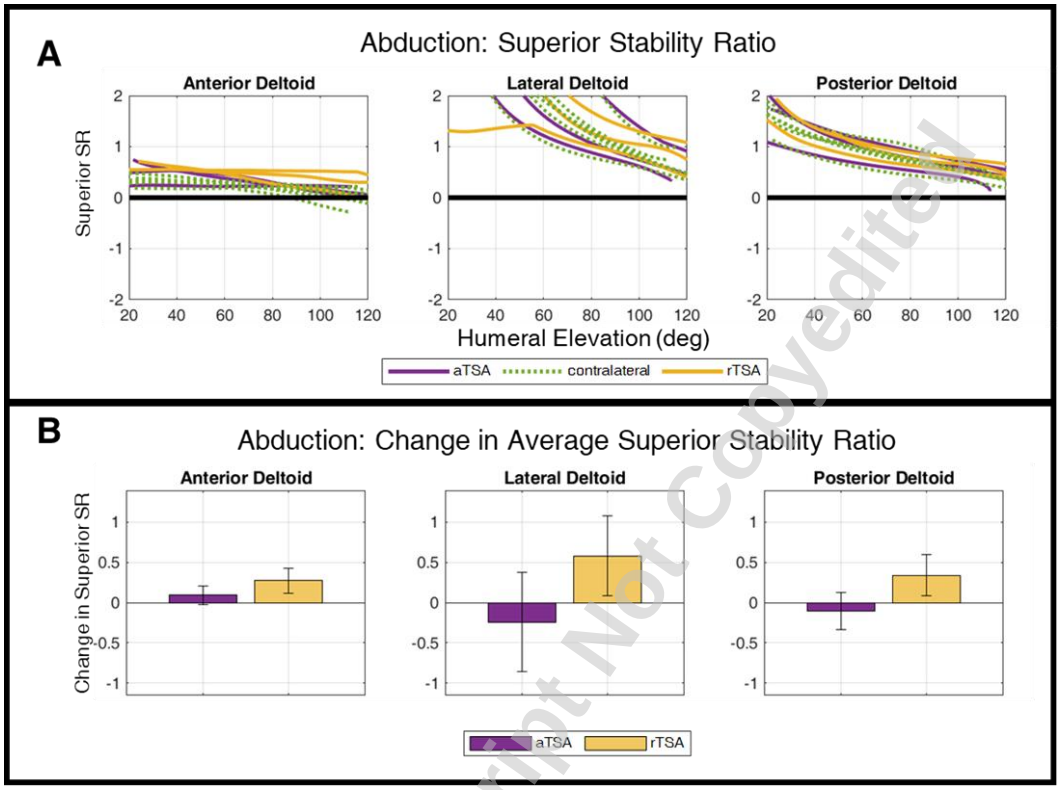

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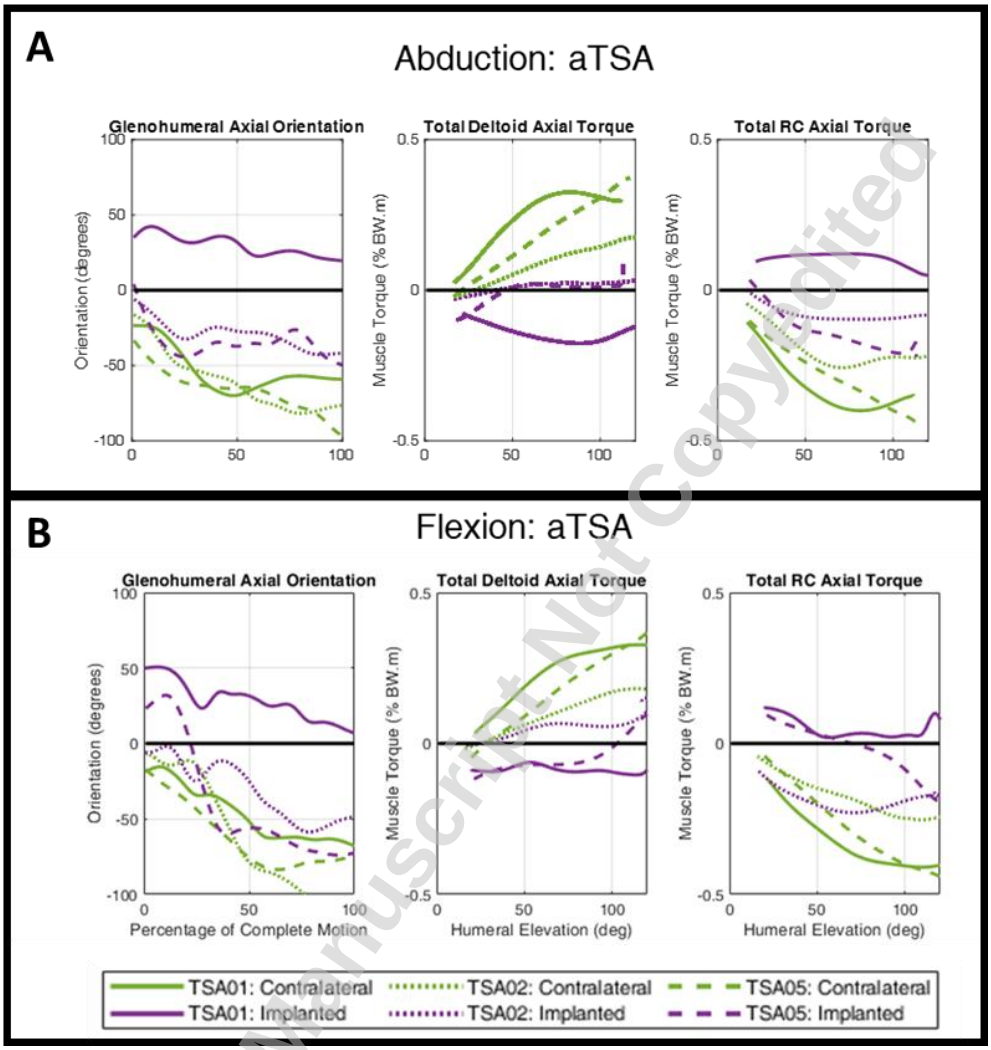

Figure 8: Glenohumeral orientation of the implanted and contralateral limbs of the three aTSA patients in the study plotted against the percentage of the complete abduction motion (left); sum of the axial rotation torque produced by the anterior, lateral, and posterior deltoid muscles plotted against humeral elevation (middle); sum of the axial rotation torque produced by the infraspinatus, subscapularis, supraspinatus, and teres minor muscles plotted against humeral elevation (right) during (A) abduction and (B) flexion. Positive values indicate internal axial orientation and internal rotation torque, respectively. Negative values indicate external axial orientation and external rotation torque, respectively. Glenohumeral orientation is defined by the 'Y' component of a YZX Euler decomposition used by Aliaj et al., (2021), and represents how much each humerus is rotated about its long axis with respect to its neutral axial orientation, as determined from CT imaging.

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Table 1: Differences in muscle group, as well as individual muscle forces between the implanted and contralateral limbs for each group (aTSA, top; rTSA, bottom) at 90 degrees of abduction. Positive values indicate increased force in the implanted limb compared to the contralateral limb.

Abduction

aTSA

Muscle Groups	Group Force Difference (%BW)	Group Force STD	Muscle	Muscle Force Difference (%BW)	Muscle Force STD
Deltoids	-0.149	2.5	Ant. Delt	-7.30	5.84
			Lat. Delt	7.03	6.39
			Post. Delt	0.12	0.02
Rotator Cuff	-9.37	3.27	Supraspinatus	-1.19	0.93
			Infraspinatus	-8.60	3.61
			Subscapularis	1.66	5.25
			Teres Min	-1.24	2.76
Others	3.89	8.49	Teres Maj	-0.01	0.03
			Pec. Maj Clav	7.05	5.79
			Pec. Maj Middle	-0.13	0.07
			Pec. Maj Inf	0.02	0.06
			Lat. Dorsi	0.10	0.03
			Coracobrachialis	-3.14	2.92

rTSA

Muscle Groups	Group Force Difference (%BW)	Group Force STD	Muscle	Muscle Force Difference (%BW)	Muscle Force STD
Deltoids	2.96	12.3	Ant. Delt	-10.32	7.83
			Lat. Delt	13.02	6.35
			Post. Delt	0.27	0.26
Rotator Cuff	-24.5	13.3	Supraspinatus**	-5.73	1.19
			Infraspinatus	-2.83	8.59
			Subscapularis	-12.55	2.90
			Teres Min	-3.38	5.97
Others	-3.27	11.5	Teres Maj	0.00	0.01
			Pec. Maj Clav	-0.78	12.79
			Pec. Maj Sup	-0.06	0.09
			Pec. Maj Inf	-0.02	0.05
			Lat. Dorsi	-0.47	0.98
			Coracobrachialis	-1.94	1.54

Key
Difference > 15%BW
5%BW < Difference < 15%BW
Negligible/Variable Difference
** Supraspinatus inactive in rTSA models

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Table 2: Differences in muscle group, as well as individual muscle forces between the native and implanted limbs for each group (aTSA, top; rTSA, bottom) at 90 degrees of flexion. Positive values indicate increased force in the implanted limb compared to the contralateral limb.

Flexion

aTSA

Muscle Groups	Group Force Difference (%BW)	Group Force STD	Muscle	Muscle Force Difference (%BW)	Muscle Force STD
Deltoids	0.339	2.13	Ant. Delt	-4.54	5.66
			Lat. Delt	1.64	7.78
			Post. Delt	2.57	4.13
Rotator Cuff	-5.52	5.47	Supraspinatus	-0.01	1.01
			Infraspinatus	-6.75	6.12
			Subscapularis	2.20	3.14
			Teres Min	-0.96	2.47
Others	-1.16	1.82	Teres Maj	0.00	0.06
			Pec. Maj Clav	-0.26	0.21
			Pec. Maj Middle	-0.04	0.06
			Pec. Maj Inf	0.01	0.04
			Lat. Dorsi	-0.14	0.35
			Coracobrachialis	-0.74	1.31

rTSA

Muscle Groups	Group Force Difference (%BW)	Group Force STD	Muscle	Muscle Force Difference (%BW)	Muscle Force STD
Deltoids	-3.67	8.26	Ant. Delt	-1.51	5.90
			Lat. Delt	-2.30	3.31
			Post. Delt	0.14	0.16
Rotator Cuff	-11.2	3.55	Supraspinatus**	-2.96	1.41
			Infraspinatus	-6.83	1.35
			Subscapularis	-0.71	3.04
			Teres Min	-0.67	1.32
Others	2.59	1.91	Teres Maj	-0.01	0.01
			Pec. Maj Clav	2.37	2.66
			Pec. Maj Sup	-0.68	0.67
			Pec. Maj Inf	-0.10	0.11
			Lat. Dorsi	-0.18	1.51
			Coracobrachialis	1.18	1.46

Key
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Negligible/Variable Difference
** Supraspinatus inactive in rTSA models

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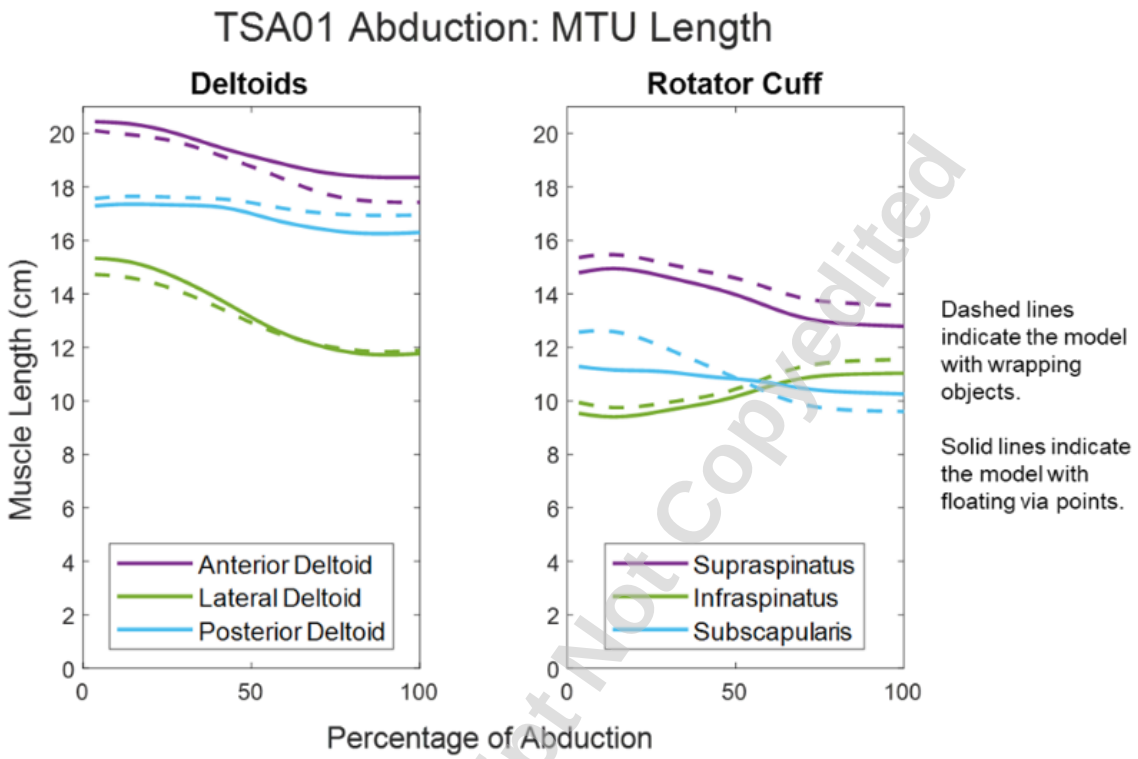
Supplemental Material



Supp. Fig. 1) Comparison of muscle tendon unit (MTU) lengths for all muscles using wrapping objects to define muscle paths and using floating via points to define muscle paths in one subject, TSA01 (aTSA) during abduction.
*all results from this study come from models using wrapping objects.

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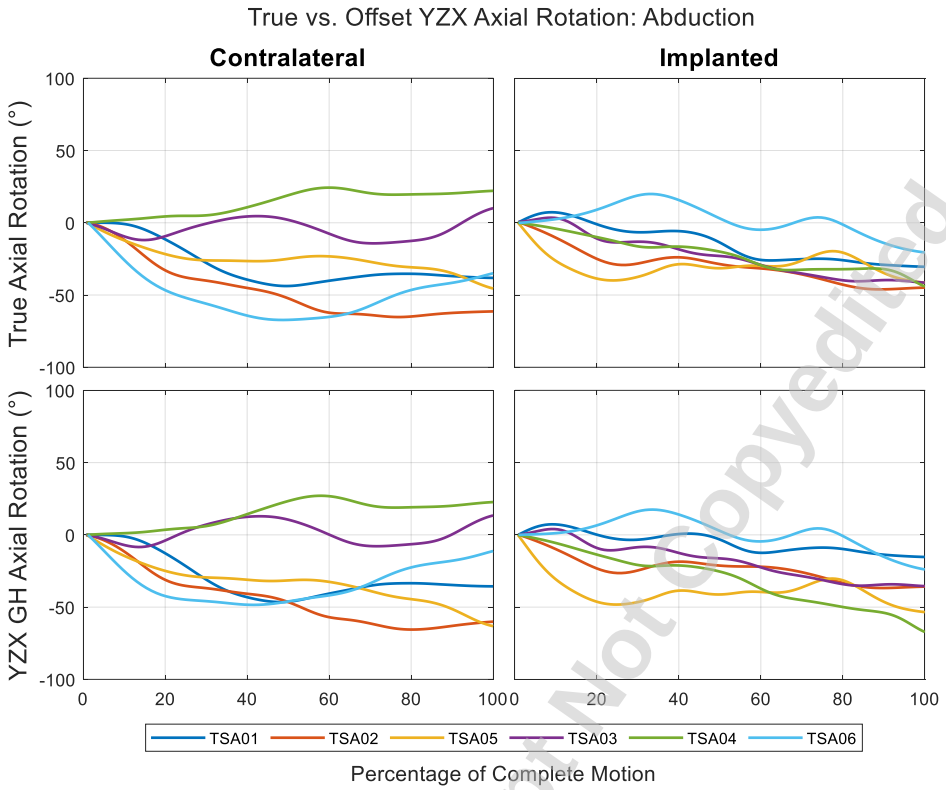

Supp. Fig. 2) Comparison of true axial rotation and axial rotation derived from YZX decompositions in both implanted and contralateral shoulders during abduction. For the purpose of comparison, all YZX decomposition axial rotation values have been offset to begin at 0° to match the starting position of all true axial rotation values.

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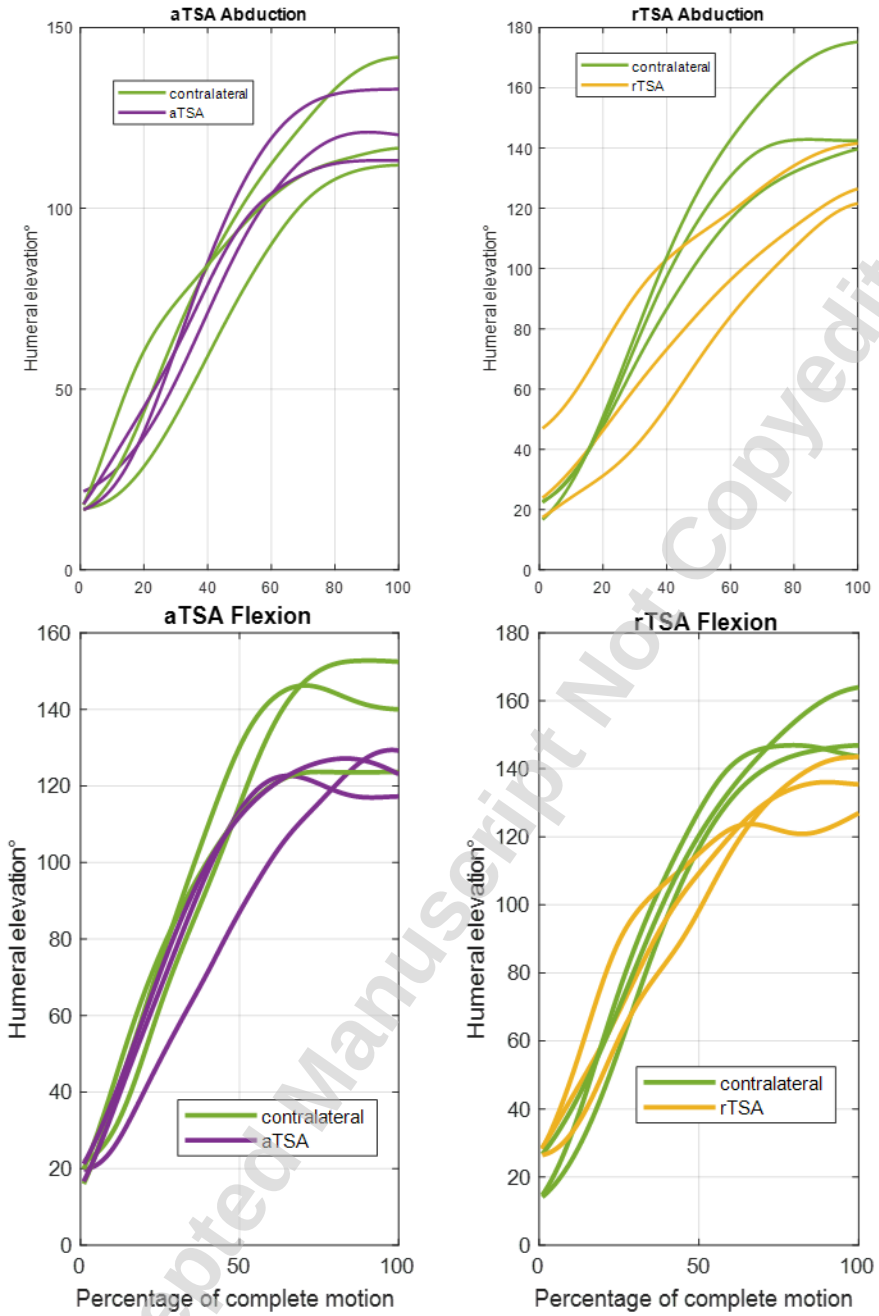

Supp. Fig. 3) Humeral elevation for all aTSA (left) and rTSA (right) patients and the associated contralateral shoulders during abduction (top) and flexion (bottom) plotted against percentage of the complete motion. Humeral elevation includes both glenohumeral and scapulothoracic elevation with respect to a neutral pose.

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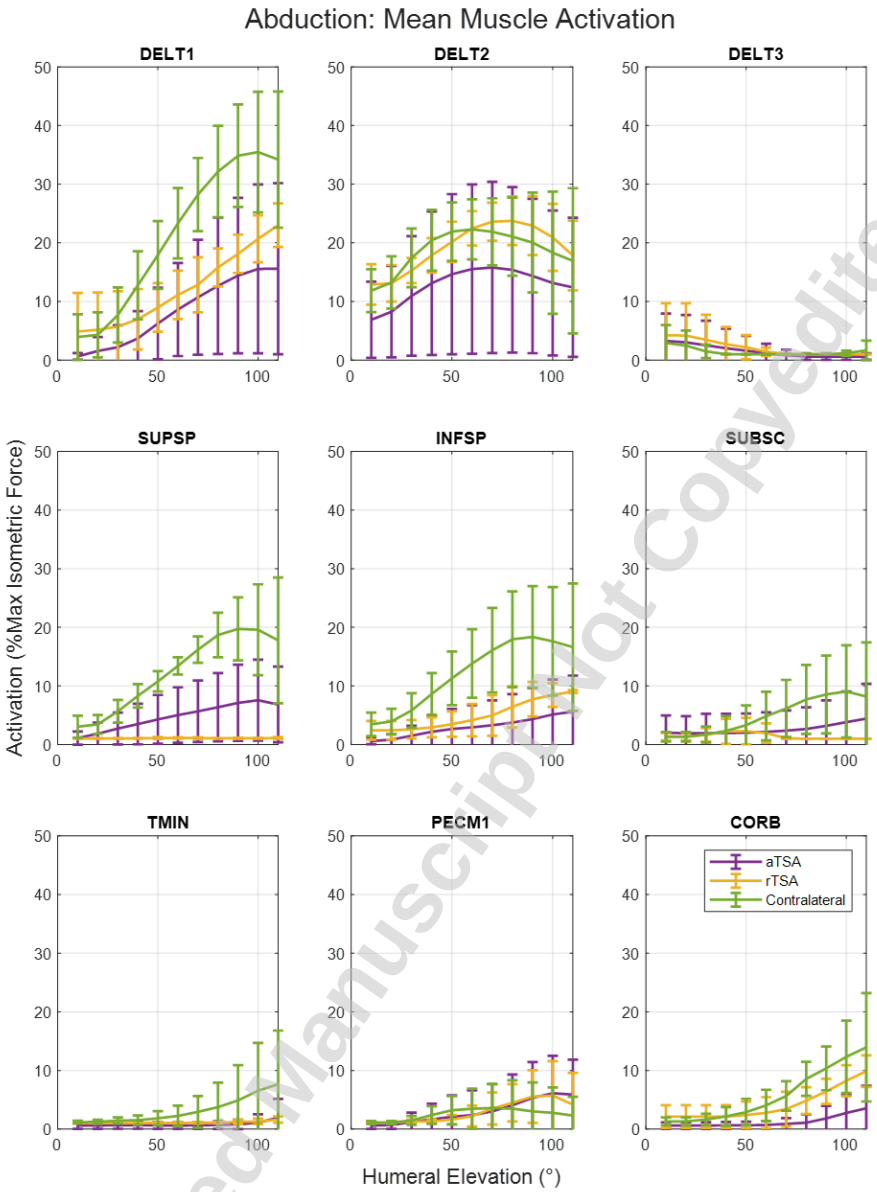

Supp. Fig. 4) Mean muscle activation (% max isometric force) of the nine most active glenohumeral muscles for all contralateral native, aTSA, and rTSA shoulders calculated at intervals of 10° humeral elevation during abduction.

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Supp. Fig. 5) Muscle forces (%BW) for the nine most active muscles for all aTSA and rTSA shoulders as well as each patient's contralateral native shoulder, calculated at 90° abduction and flexion.

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GHJRF Differences (%BW) between Implanted and Contralateral Shoulders

	90° Abduction				90° Flexion			
	Anterior	Superior	Medial	Magnitude	Anterior	Superior	Medial	Magnitude
aTSA Patients								
TSA01	-14.9	15.2	-13.7	-8.2	-1.4	9.4	-21.1	-14.0
TSA02	0.0	12.5	-3.8	0.5	2.2	7.7	10.1	9.7
TSA05	8.2	-9.4	7.6	3.0	-12.5	-19.1	-11.7	-9.9
aTSA Group Mean	-2.2	6.1	-3.3	-1.5	-3.9	-0.7	-7.6	-4.7
aTSA Group STD	11.7	13.4	10.7	5.9	7.7	16.0	16.0	12.7
rTSA Patients								
TSA03	21.8	8.8	-24.7	-16.0	5.0	12.6	-35.4	-21.7
TSA04	-11.3	11.7	-22.2	-16.5	13.6	16.2	-1.4	1.0
TSA06	-5.1	13.1	-35.7	-30.5	-5.8	-0.3	-23.1	-15.7
rTSA Group Mean	1.8	11.2	-27.5	-21.0	4.2	9.5	-20.0	-12.2
rTSA Group STD	17.6	2.2	7.2	8.3	9.7	8.7	17.2	11.8

Supp. Table 1) GH joint reaction force differences between implanted and contralateral shoulders for each individual patient, along with aTSA and rTSA group differences, at 90° abduction and flexion.

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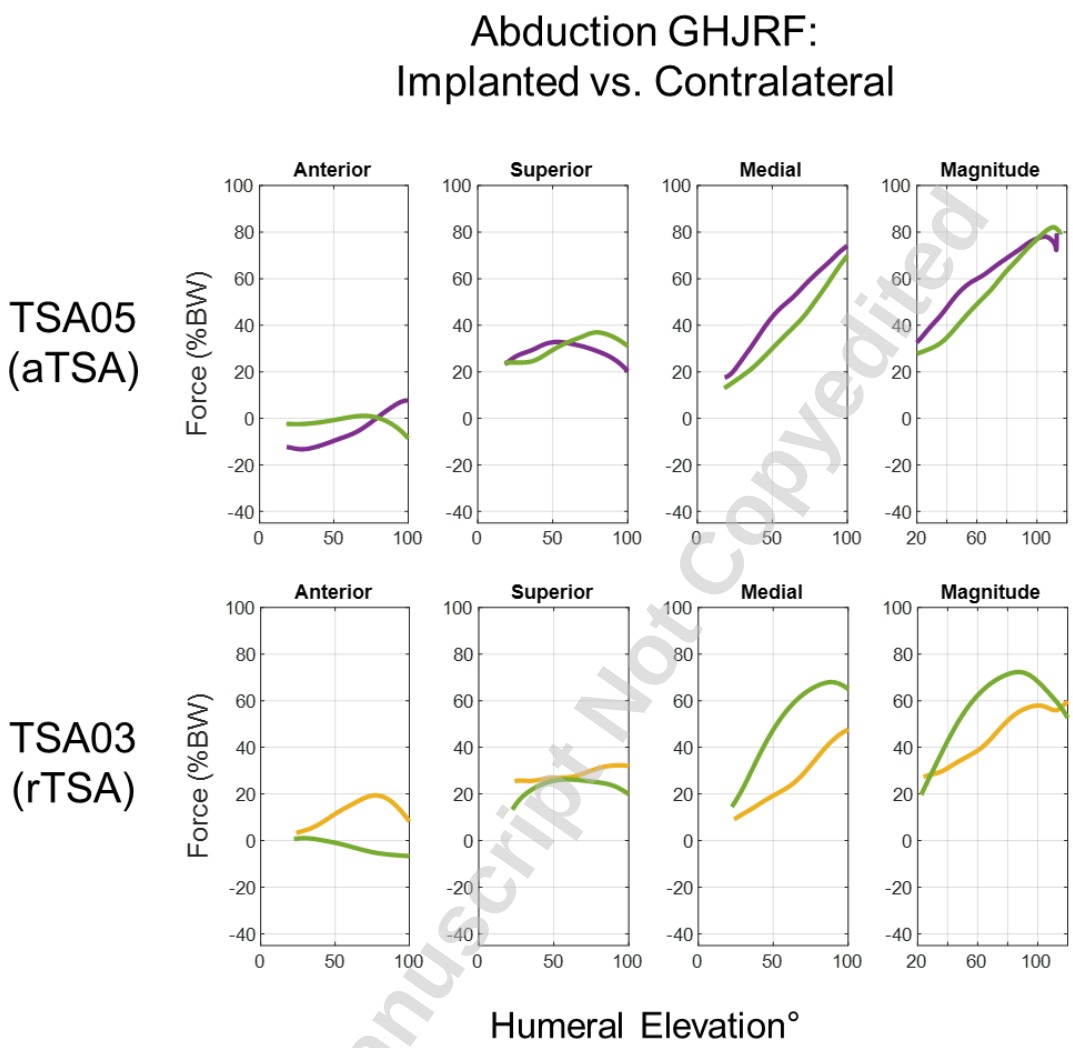

Supp. Fig. 6) Example comparison of GH joint reaction forces in the implanted and contralateral shoulders for two individual patients (TSA05: aTSA, TSA03: rTSA) during the full abduction motion.

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